

RIVERCOURSE AGCM, AB, Canada

WMO: 711020

Lat: 53.0189N

Lon: 110.1019W

Elev: 2195

StdP: 13.57

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-25.2	-20.3	-31.7	1.0	-25.0	-26.4	1.4	-20.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	82.6	63.5	79.4	62.2	76.4	61.0	67.3	77.9	65.2	75.4	63.2	72.7	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.3	94.4	73.3	61.3	87.7	70.6	59.2	81.4	68.2	33.1	78.4	31.3	75.4	29.8	72.9	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-32.5	87.8	3.2	3.2	-34.8	90.1	-36.6	92.0	-38.5	93.7	-40.8	96.1		
(5)			-32.6	71.7	3.2	2.3	-34.9	73.4	-36.8	74.7	-38.6	76.0	-41.0	77.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	9.3	8.1	21.2	36.6	51.5	59.0	62.8	60.4	52.3	38.8	21.9	7.8		
	DBStd	23.13	15.16	14.02	13.51	9.68	8.17	5.93	4.90	5.63	8.29	8.90	12.60	14.16		
	HDD50	6412	1263	1174	894	410	82	8	0	2	69	360	842	1308		
	HDD65	10672	1728	1594	1359	852	421	193	102	160	386	812	1292	1773		
	CDD50	1292	0	0	0	8	129	279	397	326	139	14	0	0		
	CDD65	74	0	0	0	0	3	13	34	19	5	0	0	0		
	CDH74	1201	0	0	0	4	117	185	415	309	166	5	0	0		
	CDH80	228	0	0	0	0	14	28	80	59	47	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	15.9	0.7	0.5	0.6	1.1	1.7	2.9	3.1	2.2	1.3	0.7	0.6	0.7		
(15) Precipitation	PrecMax	23.3	1.5	1.1	1.7	3.2	4.5	5.6	8.4	6.7	3.6	1.6	2.0	1.3		
	PrecMin	8.5	0.2	0.1	0.1	0.1	0.2	0.5	0.6	0.2	0.1	0.1	0.1	0.2		
	PrecStd	3.6	0.4	0.2	0.3	0.9	1.1	1.2	1.7	1.2	1.0	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.2	39.4	53.2	70.9	81.9	84.0	86.5	85.3	85.5	71.9	53.5	39.9		
		MCWB	35.2	35.2	41.9	48.7	56.4	62.3	67.6	64.3	61.1	53.6	43.2	34.5		
	2%	DB	37.2	35.7	46.0	64.0	77.5	79.0	81.8	81.0	78.6	62.3	45.6	35.8		
		MCWB	33.5	32.6	38.6	46.7	54.9	60.5	67.1	63.6	58.1	48.8	38.9	32.4		
	5%	DB	34.5	32.6	40.8	58.4	72.5	75.4	78.6	77.3	73.0	57.2	40.6	31.3		
		MCWB	31.5	30.2	36.1	44.0	52.8	59.2	65.2	62.7	56.5	46.2	36.0	28.5		
	10%	DB	30.6	29.1	37.3	52.7	68.2	72.0	75.3	73.8	67.7	52.3	36.8	27.0		
		MCWB	28.4	26.9	33.6	41.1	51.1	57.6	63.5	61.4	53.9	43.7	33.2	25.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	35.8	43.3	50.4	58.7	65.8	71.8	69.4	62.5	54.9	44.7	35.2		
		MCDB	39.5	39.1	51.7	68.5	75.4	77.4	82.4	79.3	80.3	69.5	51.7	39.1		
	2%	WB	33.9	33.0	39.1	47.3	56.1	63.2	68.4	66.2	59.5	51.0	40.0	32.4		
		MCDB	37.0	35.3	45.3	62.7	72.0	75.0	79.2	76.8	76.0	59.3	45.3	36.0		
	5%	WB	31.4	30.3	36.2	44.4	54.2	61.1	66.4	64.0	57.2	47.4	36.0	28.7		
		MCDB	34.2	32.6	40.6	56.1	69.3	71.4	76.6	74.2	71.0	55.7	39.5	31.2		
	10%	WB	28.3	26.9	33.8	42.0	52.3	59.1	64.4	62.1	54.8	44.5	33.5	25.0		
		MCDB	30.5	29.1	37.0	51.5	66.3	69.3	73.4	71.5	66.2	52.3	36.7	27.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.5	16.8	19.6	23.3	20.3	21.2	21.9	22.9	19.1	14.6	14.5		
		MCDBR	17.5	18.9	19.9	29.8	29.8	26.3	26.2	28.7	32.3	27.4	19.3	18.8		
		MCWBR	14.7	16.3	13.9	15.8	13.1	12.0	13.9	14.3	14.5	15.4	13.3	16.1		
		MCDBR	17.7	18.6	19.1	27.8	25.9	22.2	24.4	24.9	29.7	25.9	17.4	18.7		
	5% WB	MCWBR	15.1	16.3	13.5	15.3	12.3	11.6	13.9	13.5	14.1	15.1	12.7	16.4		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.285	0.284	0.328	0.350	0.366	0.363	0.373	0.311	0.276	0.250	0.221		
	taud		2.338	2.236	2.396	2.371	2.357	2.341	2.380	2.327	2.477	2.548	2.452	2.369		
	Ebn at Noon		245	260	290	288	285	280	279	269	277	268	243	236		
	Edh at Noon		18	27	29	34	36	37	36	35	27	20	16	14		
(44) All-Sky Solar Radiation	RadAvg		327	663	1135	1498	1842	1885	1968	1599	1131	663	324	241		
	RadStd		32	47	88	138	134	136	108	127	122	97	36	24		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page